

IGCSE

0478 / 0984 Specification map

For examination from 2023

		Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	Unit 6	Unit 7	Unit 8	Unit 9	Unit 10
1 Data representation											
1.1	Number systems	✓									
1.2	Text, sound and images	✓									
1.3	Data storage and compression	✓									
2 Data transmission											
2.1	Types and methods of data transmission		✓								
2.2	Methods of error detection		✓								
2.3	Encryption		✓								
3 Hardware											
3.1	Computer architecture			✓							
3.2	Input and output devices				✓						
3.3	Data storage and compression			✓							
3.4	Network hardware					✓					
4 Software											
4.1	Types of software and interrupts						✓				
4.2	Types of programming language, translators and integrated development environments (IDEs)						✓				
5 The internet and its uses											
5.1	The internet and the world wide web					✓					
5.2	Digital currency					✓					
5.3	Cyber security					✓					
6 Automated and emerging technologies											
6.1	Automated systems							✓			
6.2	Robotics							✓			
6.1	Artificial intelligence							✓			
7 Algorithm design and problem-solving											
7	Algorithm design and problem-solving								✓		
8 Programming											
8.1	Programming concepts									✓	
8.2	Arrays									✓	
8.3	File handling										✓
9 Databases											
9	Databases										✓
10 Boolean logic											
10	Boolean logic										✓

IGCSE 0478 / 0984 Specification map

2020-2021 Syllabus

Notes: Units 7 and 8 are designed to cover the theoretical elements of Section 2.
It is intended that the remainder of the guided learning hours are spent learning how to program.

		Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	Unit 6	Unit 7	Unit 8
1.1 Data representation									
1.1.1	Binary Systems	✓							
1.1.2	Hexadecimal	✓							
1.1.3	Data storage	✓		✓					
1.2 Communication and Internet technologies									
1.2.1	Data transmission		✓						
1.2.2	Security aspects		✓						
1.2.3	Internet principles of operation		✓						
1.3 Hardware and software									
1.3.1	Logic gates			✓					
1.3.2	Computer architecture and the fetch-execute cycle			✓					
1.3.3	Input devices				✓				
1.3.4	Output devices				✓				
1.3.5	Memory, storage devices and media			✓					
1.3.6	Operating systems					✓			
1.3.7	High-and-low-level languages and their translators					✓			
1.4 Security									
1.4.1	Safety of data					✓			
1.4.2	Firewalls, protocols and encryption					✓			
1.4.3	Online system security					✓			
1.4.4	Real-life applications					✓			
1.5 Ethics									
1.5.i	Copyright and plagiarism						✓		
1.5.ii	Software, freeware and shareware					✓			
1.5.iii	Ethical issues, hacking, cracking and malware		✓				✓		
2.1 Algorithm design and problem-solving									
2.1.1	Problem-solving and design							✓	✓
2.1.2	Pseudocode and flowcharts							✓	
2.2 Programming									
2.2.1	Programming concepts							✓	
2.2.2	Data structures; arrays							✓	
2.3 Databases									
2.3	Data types, primary keys and QBE								✓